HEALTHCARE TECHNOLOGY · STUDENT PROJECT

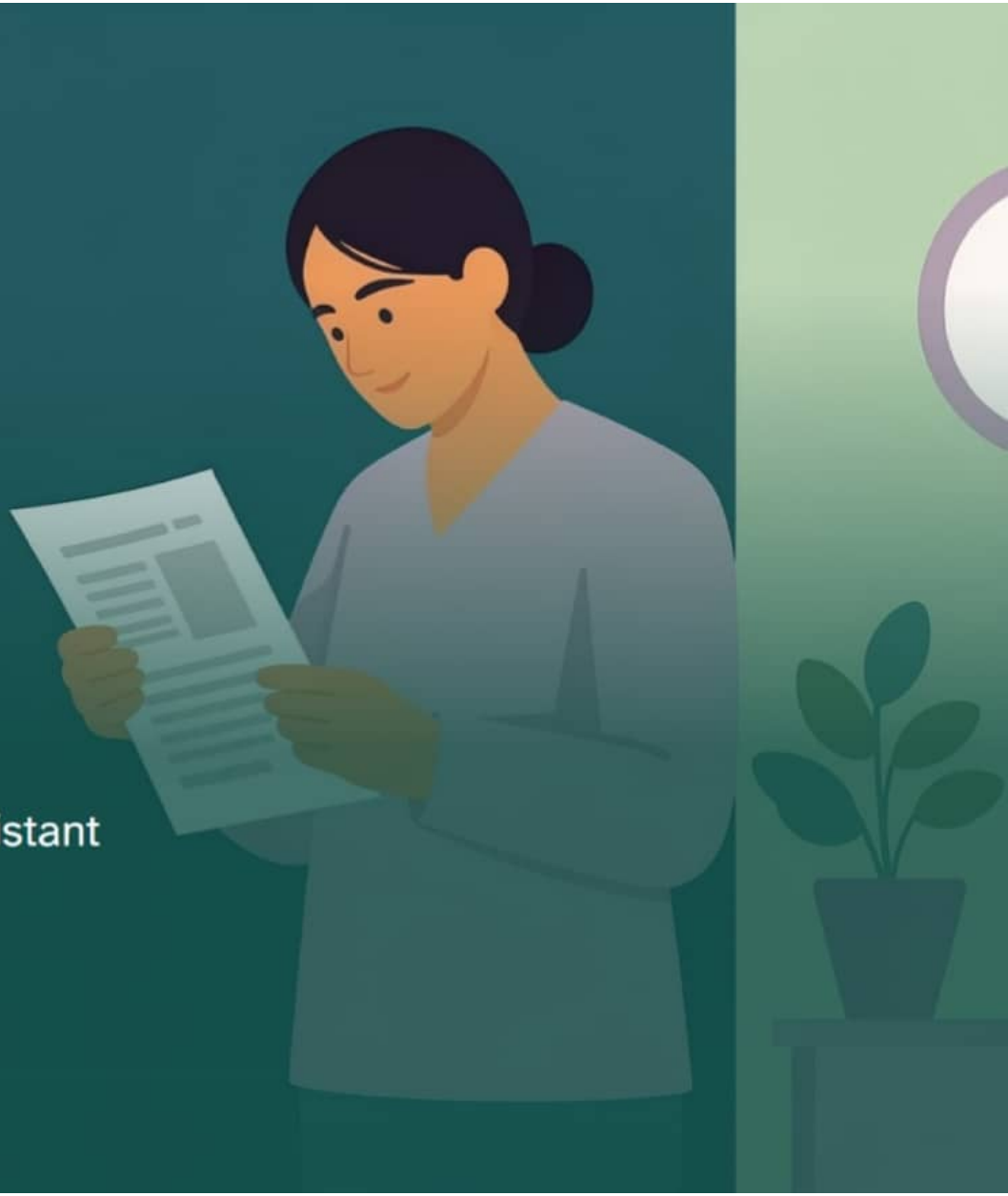
HealthSync AI

AI-powered medical report analysis and health assistant

Presented by Suryansh Shivhare, Himanshu Kashyap and Aniket

Artificial intelligence, healthcare technology and database management

Understand your health. Make informed decisions.



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THE OPPORTUNITY

Medical reports should be easier to understand

HealthSync AI helps users turn complex report values into plain-language explanations and actionable context.



AI-assisted report analysis



Explanations for medical values



Alerts for abnormal readings

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THE PATIENT PROBLEM

Patients face a translation problem

1 Medical terminology is difficult to interpret

Medical reports use terminology that is difficult for ordinary users to interpret.

2 Abnormal results are easy to miss

Abnormal results can be hard to spot across multiple reports.

3 Information stays scattered

Important information is often scattered instead of connected over time.



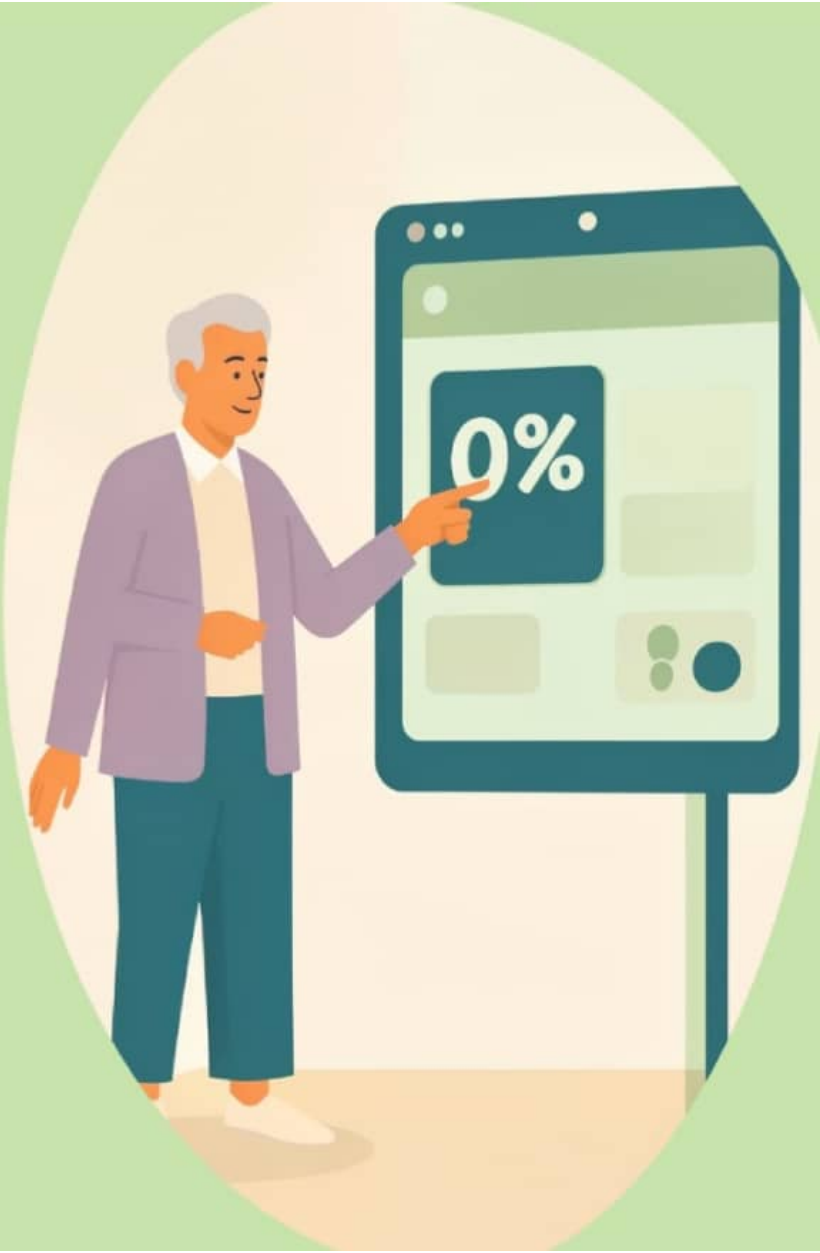
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THE GUIDED VIEW

One guided view connects the dots

HealthSync AI presents values, explanations and potential warning signs in a user-friendly interface.

01	02	03	04
Recognize the parameter	**Compare with a reference range**	**Explain the result plainly**	**Surface an alert when attention may be needed**



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THE PRODUCT JOURNEY

Designed around a clearer health journey



Simplify interpretation



Identify abnormal values



Generate understandable explanations

1. Maintain an organized report history
2. Build an accessible, interactive dashboard

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THE USER EXPERIENCE

From sign-in to insight

One path moves from access to analysis, understanding, exploration and personalization.

1. **Access**

   Sign-in, registration and demo access

2. **Analyze**

   Submit report text or a supported document

3. **Understand**

   Biomarker analysis, health alerts and explanations

4. **Explore**

   Dashboard, report history, organ insights and 3D body model

5. **Personalize**

   Dark and light themes

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IMPLEMENTATION TODAY

The current foundation is browser-first

Layer	Current role	Status
HTML5	Application structure	Current foundation
CSS3	Layout and visual design	Current foundation
JavaScript	Logic and interactions	Current foundation
Anime.js	Interface animations	Current foundation
Browser localStorage	Registration, login sessions and report-related state	Currently used

Future path: MongoDB and a backend API are intended for connected, persistent storage when implemented and tested.

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THE ANALYSIS FLOW

The analysis flow keeps the user oriented

1 User

User logs in or creates an account

2 Web Interface

User submits report text or a supported document

3 Report Analyzer

Application extracts recognized medical parameters

4 Biomarker Analysis

Values are compared with predefined reference ranges

5 Results Dashboard

Results show normal values, abnormal readings and explanations

6 Connected Database (Conditional)

Future connected storage can preserve report information for retrieval

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THE DASHBOARD

The dashboard turns analysis into a health history

- Report statistics and health summaries in one view
- Previously analyzed reports available through report history
- Organ insights connect findings to a visual body model
- Structured information prepares users for better conversations with healthcare professionals



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THE CONNECTED FUTURE

MongoDB is the path to connected persistence

MongoDB fits the project because it stores flexible, document-based records for users, report metadata, biomarker values and analysis results.

Collection	Stored fields
users	User ID, name, email and profile information
medical_reports	Report ID, user ID, type and date
biomarkers	Parameter, measured value, unit and reference range
analysis_results	Summary, flagged values and alert level

Connected persistence is the intended path when a backend is implemented and tested.



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